

CL141
ENGINEERING MECHANICS

Laboratory Manual

First Year B.Tech. (CL/ME/EE)



CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
Faculty Of Technology & Engineering
Manubhai Shivabhai Patel Department of Civil Engineering
Academic year: 2018-19

CL 141 ENGINEERING MECHANICS

Course Objective

- CO1** Understand the vector and scalar representation of forces and moments
- CO2** Identify various forces and understand effect of those forces on rigid bodies at the state of rest or motion.
- CO3** Construct free-body diagrams of rigid bodies at static equilibrium
- CO4** Comprehend mechanics associated with friction forces
- CO5** Identify and analyze the internal forces in statically determinate beams.
- CO6** Understand the distributed loads and the centroid of areas/objects.
- CO7** Understand simple dynamic variables involving kinematics, energy and momentum.

Learning Outcome

- LO1** Understand laws of mechanics and their application to engineering problems.
- LO2** Use scalar and vector analytical techniques for analyzing forces in statically determinate structures.
- LO3** Apply fundamental concepts of kinematics and kinetics of particles to the analysis of simple practical problems.
- LO4.1** Understand the fundamentals of statics and be able to apply them to simple structural problems
- LO4.2** Understand the fundamentals of dynamics and be able to apply them to simple structural problems

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List of Experiments

Sr. No.	Topic	Course Outcome	Learning Outcome
1	Law of Parallelogram of Forces	CO1, CO2	LO1, LO2
2	Law of Triangle of Forces	CO1, CO2	LO1, LO2
3	Equilibrium of Co-Planar Concurrent Force System (Universal Force Table and Polygon Law of Forces)	CO1, CO2	LO1, LO2
4	Coefficient of Static Friction	CO4	LO4.1
5	Center of Gravity of Plane Lamina	CO6	LO4.1
6	Support Reactions of a Simply Supported Beam	CO5	LO4.1
7	Study of System of Pulleys	CO3	LO1, LO2
8	Study of Fundamentals of Kinematics and Kinetics of Particles	CO7	LO3

CERTIFICATE

This is to certify that Mr./ Ms. _____
Of Division_____, Branch_____, Batch_____
Roll No._____, has satisfactorily completed his / her term work
in the subject **ENGINEERING MECHANICS (CL 141)** for the term
ending in _____20___ / 20___.

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Academic Year 2018-19

Date:

Sign of the Faculty

Head of the Department

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Sr. No.	Date	Name of the Experiment	Page No	Grade	Signature with Assessment date
1		Law of Parallelogram of Forces			
2		Law of Triangle of Forces			
3		Equilibrium of Co-Planar Concurrent Force System (Universal Force Table and Polygon Law of Forces)			
4		Coefficient of Static Friction			
5		Center of Gravity of Plane Lamina			
6		Support Reactions of a Simply Supported Beam			
7		Study of Systems of Pulleys			
8		Study of Fundamentals of Kinematics and Kinetics of Particles			

EXPERIMENT NO.: 1

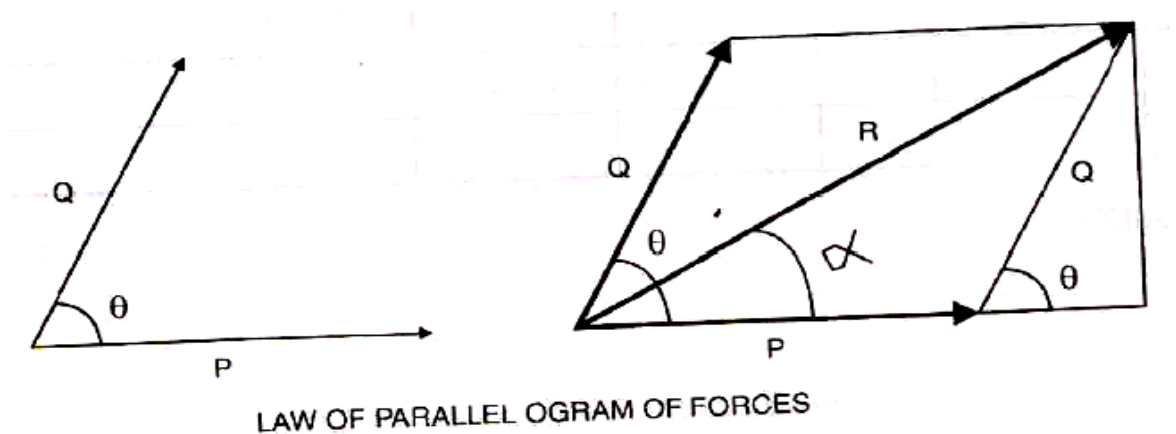
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LAW OF PARALLELOGRAM OF FORCES

Aim: To verify the law of parallelogram of forces through experiment.

Apparatus: Drawing board, thread, pan, weights, paper, scale, pulley or universal force table with slotted weights.

Theory: The law states that 'If two forces acting at a point be represented in magnitude and direction, by the two adjacent sides of a parallelogram then the diagonal of the parallelogram through the same point represents, in magnitude and direction, the resultant of the two forces.'



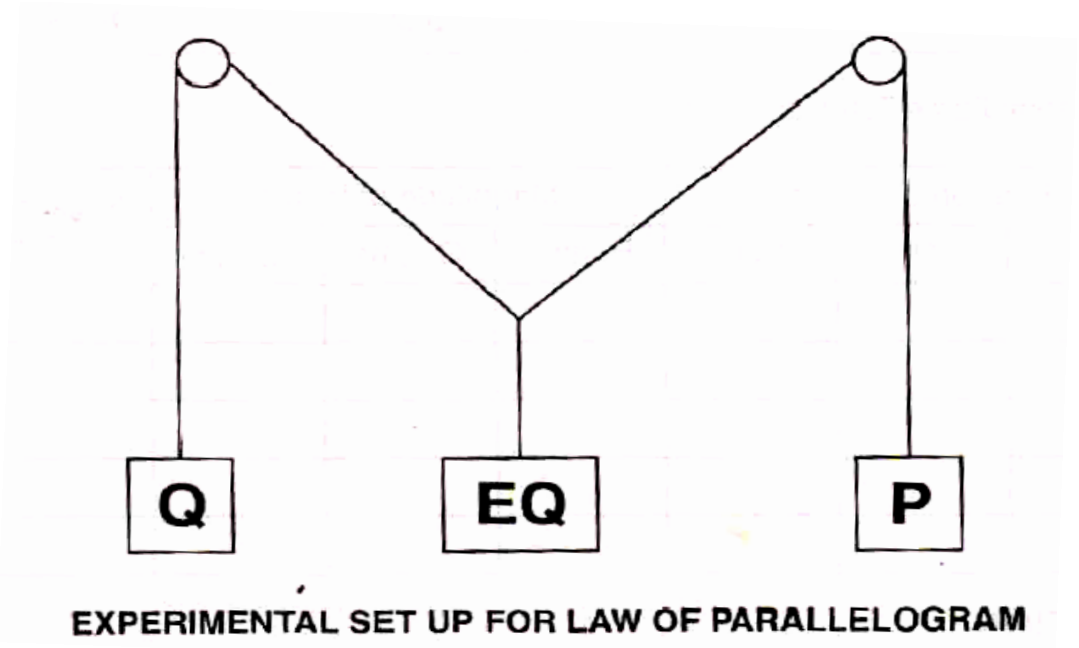
$$R = (P^2 + Q^2 + 2 PQ \cos \theta)^{1/2}$$

$$\alpha = \tan^{-1} [Q \sin \theta / (P + Q \cos \theta)]$$

Procedure: *When drawing board is used*

1. Place the drawing board in vertical position.
2. Attach the pulleys at left and right corner as shown in figure.

3. Pass the thread on these pulleys as shown in figure.
4. Attach the pan at all three ends.
5. Stick a paper on drawing board behind the thread.
6. Put some weights on each pan such that angle is not very acute or obtuse.
7. Mark the direction of each thread and also note the weights on each pan.
8. Take out the paper and join the lines.
9. Measure all the angles.
10. Consider right side force as P and left side force as Q and vertical force as Equilibrant force.
11. Transfer these lines in your manual page.
12. Select a suitable scale and hence Resultant.
13. Calculate R and α by analytical and graphical method.
14. Compare the values of R and α with graphical values.



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Observation Table:

Sr. No	Force P (gm)	Force Q (gm)	Force EQ (gm)	Angle between P & Q (θ)

Calculate R and α by following equation to verify result

$$R = (P^2 + Q^2 + 2 PQ \cos \theta)^{1/2}$$

$$\alpha = \tan^{-1} [Q \sin \theta / (P + Q \cos \theta)]$$

Calculation:

Graphically :

Scale :

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Results:

Analytical $R = \dots\dots\dots$ And $\alpha = \dots\dots\dots$

Graphical $R = \dots\dots\dots$ And $\alpha = \dots\dots\dots$

Conclusion:

Date: **Grade Obtained:** **Signature:**

EXPERIMENT NO.: ____

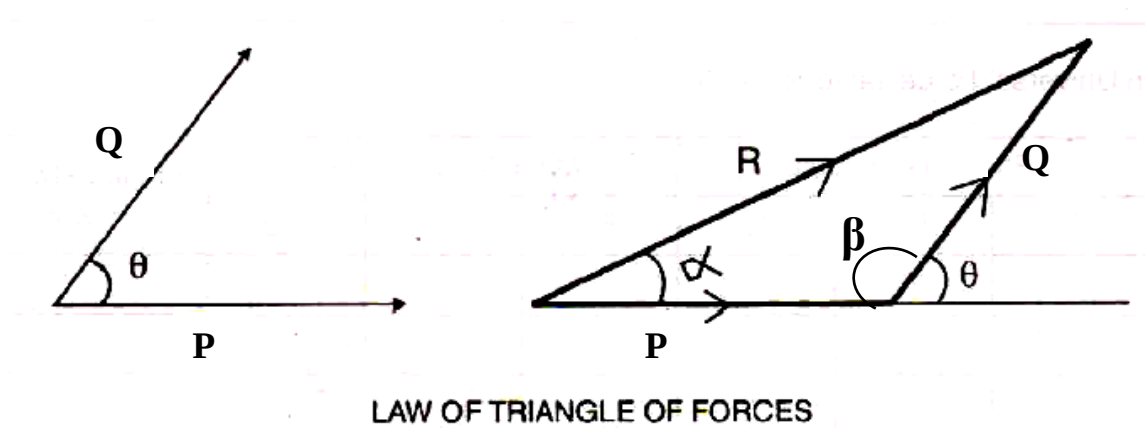
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LAW OF TRIANGLE OF FORCES

Aim: To verify the law of triangle of forces through experiment.

Apparatus: Drawing board, thread, pan, weights, paper, scale, pulley or universal force table with slotted weights.

Theory: The law states that 'If two forces acting at a point simultaneously be represented in magnitude and direction, by the two sides of a triangle take in order then the side of the triangle represents the resultant in magnitude and direction opposite in order.'



Procedure: When drawing board is used

1. Follow Steps of Experiment Law of Parallelogram
2. Consider right side force as P, and left side force as Q.
3. Select a suitable scale and draw triangle as shown in figure
4. Measure length of R and find out resultant in grams.
5. Calculate R & α by analytical method.
6. Compare the values of R & α with graphical values.

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Observation Table :

Sr. No	Force P gms	Force Q gms	Force EQ gms	Angle between P & Q (θ)	Angle between P & Q (β)

Calculate R and α by following equation to verify result

$$R = (P^2 + Q^2 - 2 PQ \cos \beta)^{1/2}$$

$$\alpha = \sin^{-1} [(Q/R) \sin \beta]$$

Calculation:

Graphically :

Scale :

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Results:

Analytical $R = \dots\dots\dots$ And $\alpha = \dots\dots\dots$

Graphical $R = \dots\dots\dots$ And $\alpha = \dots\dots\dots$

Conclusion:

Date: **Grade Obtained:** **Signature:**

EXPERIMENT NO: 3

DATE: __/__/__

**EQUILIBRIUM OF COPLANAR CONCURRENT FORCE SYSTEM
(UNIVERSAL FORCE TABLE)**

Aim: To verify the law of polygon for coplanar-concurrent forces acting on a particle in equilibrium and to find the value of unknown forces considering particle to be in equilibrium using universal force table.

Apparatus: Universal force table with four pulleys, strings, standard weights.

Theory: The state of equilibrium of a particle refers to the state of uniform velocity or rest. A particle is said to be in equilibrium under the action of forces if the vector summation of forces is zero. This experiment pertains to study the forces acting on a particle with the help of Universal force table as shown in figure 1.

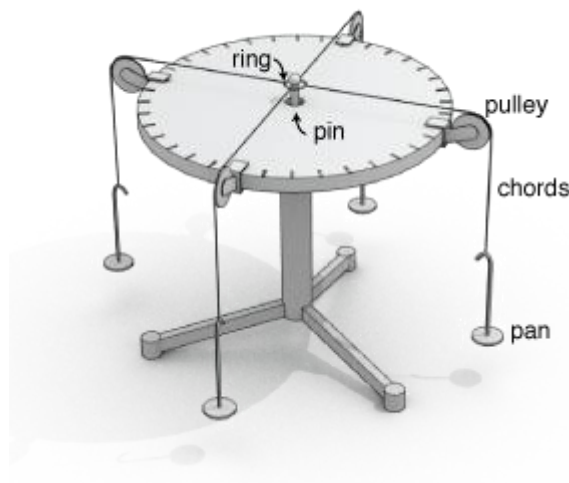


Figure 1: Universal Force Table

Brief description of the technical terms is given below:

- **Coplanar forces** All the forces of a system lie in the same plane.
- **Concurrent forces** All the forces of a system pass through a common point.
- **Equilibrium of forces** When resultant of a force system acting on a particle is zero, forces are said to be in equilibrium.

ANALYTICAL METHOD;

If $P_1, P_2, P_3, P_4,$ & P_5 are five forces acting on a particle simultaneously on a horizontal plane at an inclination of $\theta_1, \theta_2, \theta_3, \theta_4$ and θ_5 with positive X-axis measured in anticlockwise direction then the magnitude of the resultant is given by

$$R^2 = (\Sigma F_x^2 + \Sigma F_y^2)$$

Where $\Sigma F_x = F_1 \cos\theta_1 + F_2 \cos\theta_2 + F_3 \cos\theta_3 + F_4 \cos\theta_4 + F_5 \cos\theta_5$

(ΣF_x is the components of all forces along positive X-axis.)

And $\Sigma F_y = F_1 \sin\theta_1 + F_2 \sin\theta_2 + F_3 \sin\theta_3 + F_4 \sin\theta_4 + F_5 \sin\theta_5$

(ΣF_y is the components of all forces along positive Y-axis.) and its directions is given by

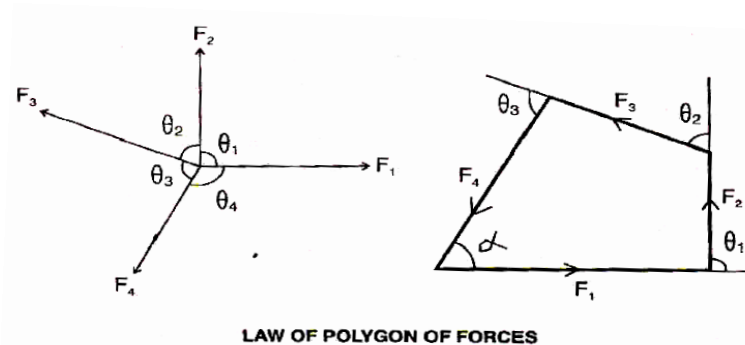
$$\theta = \tan^{-1} (\Sigma F_x / \Sigma F_y)$$

If the forces are in equilibrium the value of the resultant (R) will be zero.

This method of finding the resultant is called Resolution of forces

GRAPHICAL METHOD:

Law of polygon is employed to find the value of unknown forces graphically. Resultant of more than two coplanar concurrent forces can be found with the help of this law and is stated as “When more than two coplanar concurrent forces acting at a point are represented by the sides of a polygon taken in order, in direction and magnitude, the closing line of polygon taken in order, in direction and magnitude, the closing line of polygon, taken in opposite order, represents the resultant in direction and magnitude.” Thus polygon law of forces follows graphical method of finding the resultant of given forces.



Procedure:

1. Level the force table with the help of spirit level and adjusting foot screw.
2. Apply weights and / or adjust pulleys such that the Centre of knot coincides with central pivot. Note down the angle made by strings on graduated circular scale and the value of weights.
3. Draw spaces diagram by drawing the angles as measured on forces and show the respective forces, give Bow's notations and draw force (vector) diagram with suitable scale to solve the problem graphically. The closing line of first and last point gives the error incurred due to manual observations and friction in the apparatus. Error is found by following the procedure of resolution of forces.
4. Apply four known weights and one unknown weights, repeat the steps 2 & 3 and find the value of unknown weights analytically as well as graphically assuming the system the system with zero error.

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- Assumptions:**
1. Pulleys are assumed to be frictionless.
 2. Self-weight of the string is neglected.

- Precaution:**
1. Strings should be free of knots.
 2. Rotations of pulley should be smooth.

Observation Table:

Sr No.	Magnitude of Forces (N)				Anti-clock wise angle w.r.t +ve X- axis (degree)				ΣF_x (N)	ΣF_y (N)	Resultant 'R' (N)	
	F ₁	F ₂	F ₃	F ₄	θ_1	θ_2	θ_3	θ_4			Analytically	Graphically

Calculation:

For Analytical Method :

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For Graphical Method :

Scale :

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Results:

Analytical Resultant =

Graphical Resultant =

Conclusion:

Date: **Grade Obtained:** **Signature:**

EXPERIMENT NO:4

DATE: __/__/__

DETERMINATION OF CO-EFFICIENT OF STATIC FRICTION

Aim: To determine the coefficient of static friction between two surfaces, i.e., glass on glass, wood on glass and metal on glass.

Apparatus: An adjustable inclined glass plane with pulley at one end, wooden block having glass at bottom surface, wooden block having wood at bottom surface, wooden block having metal at bottom surface, inextensible string, pan and standard weights.

Theory:

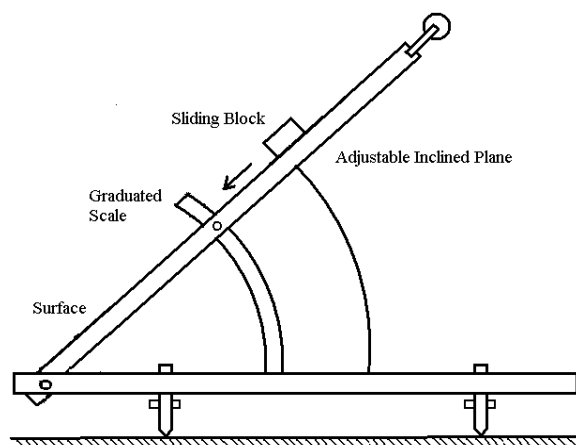


Fig. 4.1: Apparatus Used to Find Coefficient of Static Friction

- **Frictional Force (F):** “When a body moves or tends to move over another body, a force opposing the motion develops at the contact surfaces. This force which opposes the movement or the tendency of movement is called *frictional force* or simply *friction*”. If the contact surfaces are perfectly smooth, there is no friction in machines, friction is both liability and an assist, where it causes loss of power and / or wear it is undesirable. On the other hand, friction is essential for various holding and fastening devices, brakes, belt drives etc.
- **Limiting Friction Force (F_m):** The maximum friction force developed

between two surfaces when the body just starts to move in the direction of effort. I.e. Max. Friction force developed in impending condition.

-When $F < F_m$: Block is in equilibrium & Friction Force developed = F

-When $F = F_m$: Block is in impending condition & Friction Force developed = $F_m = \mu_s * N$

-When $F > F_m$: Block is in motion & Friction Force developed = $F_k = \mu_k * N$

- **COEFFICIENT OF STATIC FRICTION (μ_s):** The ratio of limiting friction force (F_m) to the normal Reaction (N) which are generated between two surfaces.

$$\therefore \mu_s = F_m / N$$

- **ANGLE OF STATIC FRICTION (Φ_s):** Angle made by the resultant (R) of normal reaction (N) and the friction force (F_m) to the normal reaction (N), when the body is in impending condition is called angle of static friction.

$$\tan \Phi_s = F_m / N$$

-Relation between angle of Static Friction and Coefficient of Static Friction:

$$\mu_s = \tan (\Phi_s)$$

- **ANGLE OF REPOSE (θ):** The angle of an inclination with horizontal at which the block just slides down due to its self-weight is called angle of Repose.

OR

The maximum inclination of plane at which the body can remain in equilibrium over the plane entirely by the assistance of friction is called the angle of repose.

(It will be seen that angle of static friction and angle of repose are equal.)

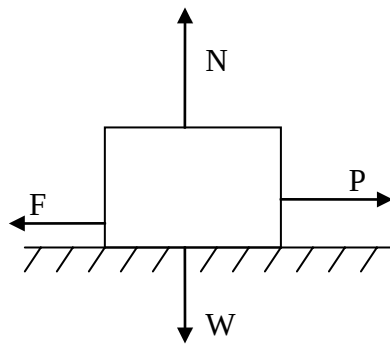


Fig. 4.2 Block Resting on Horizontal Surface

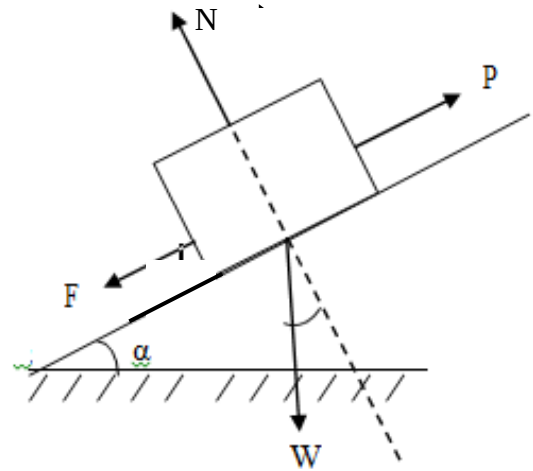


Fig. 4.3 Block Resting on Inclined Surface

- **COEFFICIENT OF STATIC FRICTION (μ_s):**

For Horizontal Surfaces:

$$\mu_s = F_m/N = P/W$$

For Inclined Surfaces:

$$\mu_s = F_m/N = (P - W \sin \alpha) / W \cos \alpha$$

Where,

μ_s = Coefficient of Static friction

W = Load

P = Effort applied

α = Inclination of plane with the horizontal

F_m = Limiting Friction force

N = Normal Reaction

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- Procedure:**
1. Set the angle of inclination of the working plane to horizontal glass surface.
 2. Place the given wooden block having glass bottom surface (W_1) on the surface of apparatus and place known weights (W_2) in this block. Here, the contact surfaces are glass and glass.
 3. Connect the block to effort pan (P_1) passing through a frictionless pulley.
 4. Apply effort (P_2) in small increments such that the block impends.
 5. Note the value of load and effort and calculate the coefficient of friction between the surfaces.
 6. Take two observations for the same block and calculate the average coefficient of friction.
 7. Also measure the angle of repose by keeping the same block on the surface and gradually tilting the incline such that the block starts slipping down, corresponding angle with the horizontal is the angle of repose. Note down that angle.
 8. Repeat steps 1 to 7 for other blocks having wooden bottom surface and metal bottom surface.

- Assumptions:**
1. Pulleys are assumed to be frictionless.
 2. Self-weight of the string is neglected.

- Precaution:**
1. Strings should be free of knots.
 2. Rotations of pulley should be smooth.

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Observations:

Self weight of the effort pan = $P_1 = 25.5 \text{ g} = \underline{\hspace{2cm}} \text{ N}$

Self weight of the Glass bottom Block = $W_1 = 242.5 \text{ g} = \underline{\hspace{2cm}} \text{ N}$

Self weight of the Wooden bottom Block = $W_1 = 132 \text{ g} = \underline{\hspace{2cm}} \text{ N}$

Self weight of the Metal bottom Block = $W_1 = 196 \text{ g} = \underline{\hspace{2cm}} \text{ N}$

Observation Table:

Sr. No.	Surface in Contact	Reading No.	Angle of Inclination of plane with Horizontal (α) in degree	Additional Load Applied in Block (W_2) (N)	Effort Required in Pan (P_2) (N)	Total Load moved ($W=W_1+W_2$) (N)	Total Effort applied ($P=P_1+P_2$) (N)	Co- efficient of Friction (μ_s)	Avg. Co- efficient of Friction (μ_s)	Angle of Repose (θ)
1.	Glass on Glass	1.								
		2.								
2.	Wood on Glass	1.								
		2.								
3.	Metal on Glass	1.								
		2.								

Calculation:

Glass on glass surface:

Wood on glass surface:

Metal on glass surface:

Conclusion:

Date: **Grade Obtained:** **Signature:**

EXPERIMENT NO:5

DATE: __/__/__

CENTER OF GRAVITY OF PLANE LAMINA

Aim:

To determine the center of gravity of different shaped plane lamina with uniform thickness.

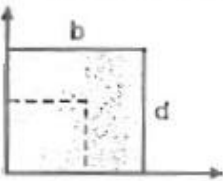
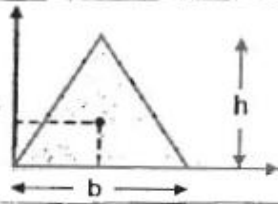
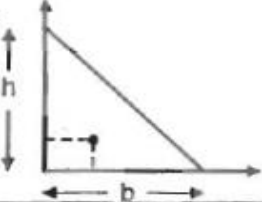
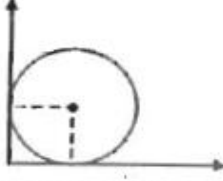
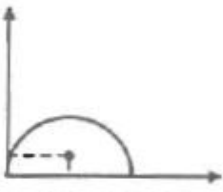
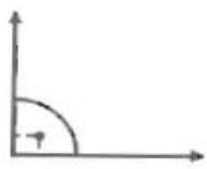
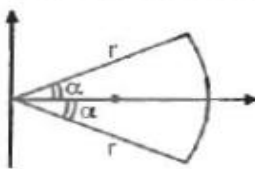
Apparatus:

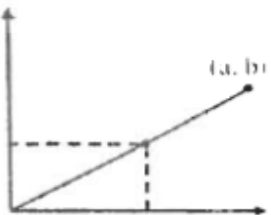
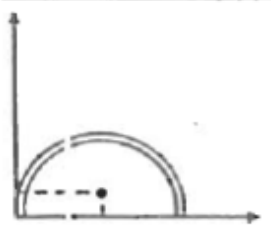
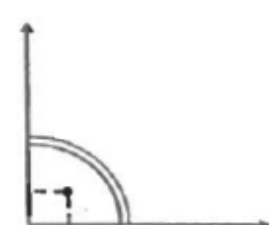
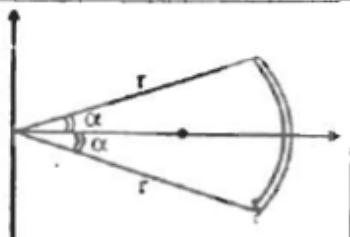
Plumb bob, white sheet and different shaped plane lamina of uniform thickness.

Theory:

- **Center of Gravity:** The center of gravity of a body is the point where the entire weight of body is assumed to be concentrated and is attracted towards the center of earth. The center of gravity does not depend on orientation of the body. Whenever the body is freely suspended, its center of gravity always remains below the line of suspension. Everybody has one and only one center of gravity.
- **Centroid:** It is defined as a point about which the entire line, area, or volume of a body is assumed to be concentrated. It is used for calculating the center of gravity of figures where weight factor does not come into picture.
- **Center of Mass:** It is defined as the point about where the entire mass of the body is assumed to be concentrated. If a body is homogenous and isotropic in nature, center of gravity and center of mass would be same.
- **Axis of Symmetry:** It is defined as the axis about which if the given figure is folded, then one portion exactly superimposes the other portion.

Table 5.1: For Centroid & Centre of gravities of basic figure:

Shape	Length /Area /Volume	$\bar{x}$	$\bar{y}$
1. Rectangle 	$A = bd$	$\frac{b}{2}$	$\frac{d}{2}$
2. Isosceles triangle 	$A = \frac{1}{2}bh$	$\frac{b}{2}$	$\frac{h}{3}$
3. Right angled triangle 	$A = \frac{1}{2}bh$	$\frac{b}{3}$	$\frac{h}{3}$
4. Circle 	$A = \pi r^2$	r	r
5. Semicircle 	$A = \frac{\pi r^2}{2}$	r	$\frac{4r}{3\pi}$
6. Quarter circle 	$A = \frac{\pi r^2}{4}$	$\frac{4r}{3\pi}$	$\frac{4r}{3\pi}$
7. Circular sector : 	$A = r^2\alpha$	$\frac{2r \sin \alpha}{3\alpha}$	0

8. Line		$l = \sqrt{a^2 + b^2}$	$\frac{a}{2}$	$\frac{b}{2}$
9. Semicircular arc :		$l = \pi r$	r	$\frac{2r}{\pi}$
10. Quarter circular arc :		$l = \frac{\pi r}{2}$	$\frac{2r}{\pi}$	$\frac{2r}{\pi}$
11. Segment of an arc :		$l = 2r\alpha$	$\frac{r \sin \alpha}{\alpha}$	0

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Procedure:

1. Fix a white paper exactly as of the size of the plane lamina.
2. Suspend the given lamina by a thread from one of the points given at its periphery.
3. Draw the vertical line on the area of lamina through the line of suspension of the thread.
4. Suspend the lamina again from another point given on the periphery and draw the line through the line of suspension of the thread.
5. Check further suspending the lamina from third point of lamina whether the line of suspension thread co-insides the same intersection point. Repeat the procedure for all the points.
6. The point of intersection of these lines gives the center of gravity of that lamina by experimentally.
7. Also calculate the center of gravity of that plane lamina by analytically.

Observation Table:

Axis of References through point 'O': _____

Sr. No.	Types of Lamina	Experimentally/Graphically		Analytically	
		$\bar{X}$ (mm)	$\bar{Y}$ (mm)	$\bar{X}$ (mm)	$\bar{Y}$ (mm)
1.	Triangle				
2.	I – Section				
3.	Composite Section				

Triangle:

Experimentally

Analytically:

I – Section

Experimentally

Analytically:

Composite

Experimentally

Analytically:

Conclusion:

Date: **Grade Obtained:** **Signature:**

EXPERIMENT NO: 6

DATE: ____/____/____

SUPPORT REACTIONS OF A SIMPLY SUPPORTED BEAM

Aim: To verify the conditions of equilibrium of a rigid body under the action of coplanar non-concurrent, parallel force system with the help of a simply supported beam.

Apparatus: Graduated beam, standard weights and hooks.

Theory:

- **Parallel Force System:** Forces which are acting parallel to each other in direction are called parallel force system.
- **Non-concurrent Force System:** Forces whose line of action not meets at a single point are called non-concurrent force system.
- **Beam :** is a structural member having cross-sectional dimensions very smaller than its length and is subjected to transverse loads (load acting perpendicular to the longitudinal axis of the beam).

Beam is an example of Coplanar Non-concurrent Force System under the equilibrium condition.

Analytical Conditions of Equilibrium for Coplanar Non-concurrent Force System:

1. Algebraic sum of forces along X-axis must be zero. i.e. $\Sigma H=0$.
2. Algebraic sum of forces along Y-axis must be zero. i.e. $\Sigma V=0$.
3. Algebraic sum of moments of forces about any point must be zero . i.e. $\Sigma M=0$.

Graphical Conditions of Equilibrium:

1. Force polygon should close which indicates that there is no motion of translation.
2. Funicular polygon should close which indicates that there is no motion of rotation.

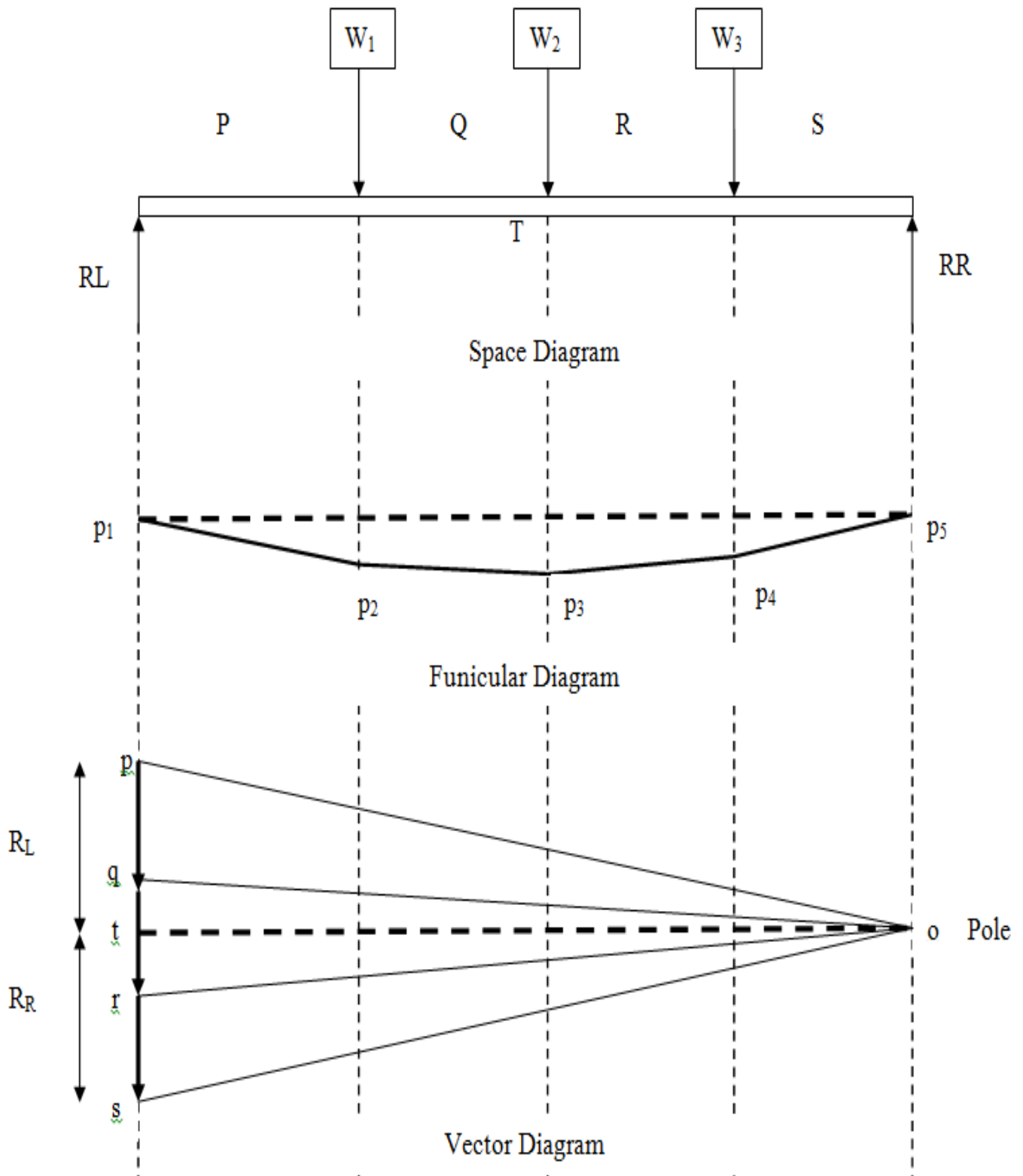
CALCULATION OF SUPPORT REACTION:

• **ANALYTICAL METHOD:**

1. Apply $\Sigma M @$ any one support = 0 for finding the vertical support reaction at one support.
2. Apply $\Sigma V = 0$ for finding the remaining vertical support reaction.
3. Apply $\Sigma H = 0$ for finding the horizontal support reaction.

• **GRAPHICAL METHOD:**

1. Draw the space diagram of the beam, and name all the loads according to Bow's notation.
2. Construct Vector diagram. For that select some suitable point 'p' and draw 'pq' parallel and equal to load W_1 to some scale. Similarly through 'q' draw 'qr' parallel and equal to load W_2 and through 'r' draw 'rs' parallel and equal to load W_3 to the scale. Select any suitable point 'o' and join 'op', 'oq', 'or' and 'os'.
3. Construct funicular diagram. For that select any point 'p₁' on the line of action of the reaction 'RL'. Through 'p₁' draw 'p₁p₂' parallel to 'op' intersecting the line of action of the load W_1 at p₂. Similarly, draw 'p₂p₃', 'p₃p₄' and 'p₄p₅' parallel to 'pq', 'or' and 'os' respectively. Join p₁ with p₅ and through o draw a line 'ot' parallel to this line. Now the lengths 'tp' and 'st', in the vector diagram, given the magnitude of the reaction RL and RR respectively to the scale.



- Procedure:**
1. Place the beam in its position on a selected value of the span.
 2. Note down the initial readings shown by the spring balance at reaction points. Here, readings may be observed zero. Because the setup of the dial gauge of apparatus has been done such that, the weight of the beam will not to be observed.
 3. Suspend several weights on the beam and note the magnitude of the loads and their distance from the left support.
 4. Observe the readings shown by spring balance at the reaction points.
 5. Calculate the experimental value of the support reactions by deducting initial readings from the final reading.
 6. Verify the condition of equilibrium for the coplanar non-concurrent force system.
 7. Calculate the values of the support reactions by analytically as well as graphically.
- Assumptions:** The setup of the dial gauge of apparatus has been done such that, the self-weight of the beam is not to be considered in calculations.

Observation Table:

To verify the condition of equilibrium for coplanar non- concurrent force system													
Sr. No.	Applied Loads (N)			Observed Reactions by Experiment (N)				Distance of Loads from Left Support (From 'A') (mm)	Length of the Beam L (mm)	ΣH (N)	ΣV (N)	ΣM_A (N.m)	
	W1	W2	W3	R _L		R _R		X ₁	X ₂	X ₃			
				Initial	Final		Initial	Final					
1													
2													
3													

Calculation:

Calculation of support reaction by analytically:

Calculation of support reaction by graphically:

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Result Table:

Sr. No	Experimentally (N)		Analytically (N)		Graphically (N)	
	R _L	R _R	R _L	R _R	R _L	R _R
1.						
2.						
3.						

Conclusion:

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EXPERIMENT NO: 7

DATE: __/__/__

STUDY OF SYSTEMS OF PULLEYS

Aim: To study the systems of pulleys and draw the free body diagram of the system.

Theory: **PULLEY:**



A **pulley** is simply a collection of one or more wheels over which you loop a rope to make it easier to lift things. It is an example of simple machine. When a pulley is hinged to a fixed support, it is a **Fixed Pulley**. It does not mean that it cannot move. It changes the direction of the force on a rope or belt that moves along its circumference. It is free to rotate around the fixed pivot. The pulley which is made free to move up and down along with the load, it is called a **Movable Pulley**. It can rotate like fixed pulley and is supported by two parts of the same rope and has a mechanical advantage of two. Hence, we can say that a single fixed pulley can be used to change only the direction of the applied force whereas a movable pulley can be used to decrease the input force required.

➤ **Function Of Pulley:**

The function of a pulley is to lift heavy objects by changing the direction of the force on a flexible cable. It also consist a wheel with a groove in its outer edge and an axle. With the help of ropes, chains or cords, a pulley makes moving objects upward easier. Pulley reduces the force given to get the work done. The direction of the force required to lift an object is changed from pushing up to pulling down while using a pulley system. Pulleys are manufactured to make hefty work more manageable and are

classified as movable, fixed or combined. Multiple pulleys can also decrease the force needed to lift or pull the weight. Pulley also functions to mainly distribute the weight evenly which sort of makes the elevator lighter. The pulley serves in many different industries for lifting loads, applying forces or transmitting power.

➤ **Application Of Pulley:**

Pulley is a simple machine with varied complex uses. Pulleys are used to change the direction of an applied force, transmit rotational motion, or realize a mechanical advantage in either a linear or rotational system of motion. The advantage is that in exchange for very little effort, difficult tasks can be completed. Pulley acts as an asset for the common user so as to create motion transfer with the least amount of effort.

Pulleys have miscellaneous uses such as lift loads, apply forces, and to transmit power. It is designed to support movement of a cable or belt along its circumference.

Other Applications:

- Pulley used for drinking water – Use to lift water from the well by making use of pulley
- Pulley in Flag Raising – Use to raise the flag to its pole.
- Pulleys in Elevators - Used in lifting the cars where counterweight is used to balance the weight of the car.
- Pulley in Vehicles - Pulleys is used in vehicles to increase their power and make them run more smoothly.

➤ **Advantages & Disadvantages of Pulley:**

Advantages Of Pulley:

The pulley system is a simple device that uses a rope attached around a wheel to lift heavy objects. Pulleys are simple to make and can be used at a distance from each other

- It helps in applying force in any direction
- Pulley systems rely on the relationship between load and effort.
- Useful for getting the drive action to happen in awkward places
- Mechanical advantage is defined as the ratio of load to effort.
- A movable pulley is a pulley that moves with the load.
- Increases Lifting Distance

Disadvantages of Pulley:

Pulls the rope a long distance to move the load a relatively short distance

- There will be more loss of energy to friction with more pulleys.
- It works by friction and can slip

➤ **Working Principle of Pulley:**

When one end of the rope is pulled downwards, the load on the other end of the rope is pulled upward. Therefore, the direction of force is changed from downwards to upwards. Pulley can work on one, two or four wheels.

One wheel

If you have a single wheel and a rope, a pulley helps you reverse the direction of your lifting force. So, as shown in figure 1, you pull the rope down to lift the weight up. If you want to lift something that weights 100kg, you have to pull down with a force equivalent to 100kg, which is 1000N. If you want to raise the weight 1m into the air, you have to pull the loose end of the rope a total distance of 1m at the other end.

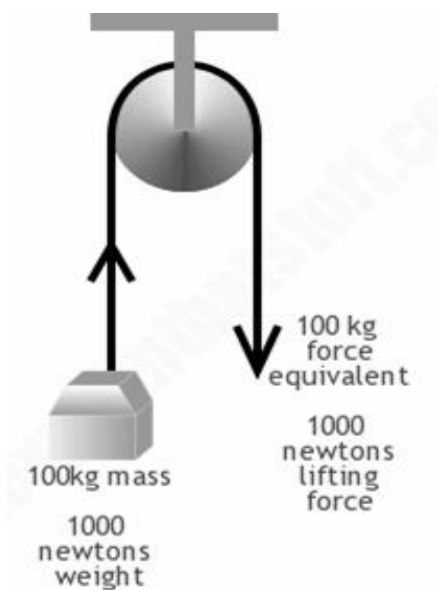


Figure 1
One wheel

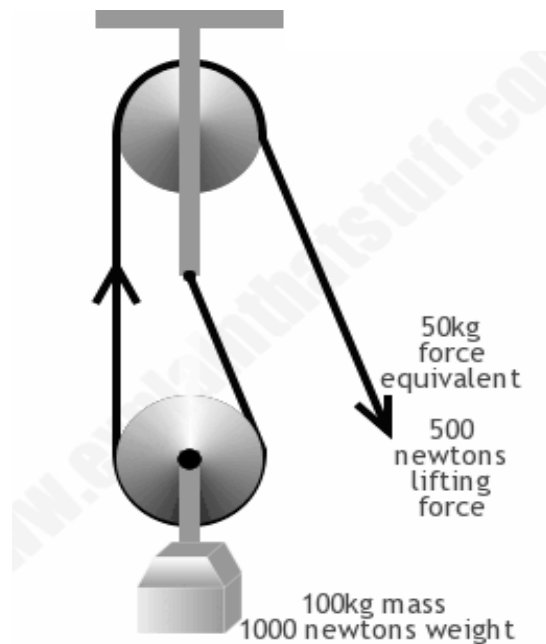


Figure 2
Two wheels

With one wheel, a pulley simply reverses the direction of the force you apply. It doesn't alter the force in any other way.

Two wheels

If you add more wheels, and loop the rope around them, you can reduce the effort you need to lift the weight. Suppose you have two wheels and a rope looped around them, as in the figure 2. The 100kg mass (1000 newton weight) is now effectively supported by two sections of the same rope instead of just one and this means you can lift it by pulling with a force of just 500 newton—half as much! That's why we say a pulley with two wheels, and the rope wrapped around it this way, gives a mechanical advantage of two.

Four wheels

Suppose you have two wheels and a rope looped around them, as in the figure 3. The 100kg mass (1000 newton weight) is now hanging from four sections of rope. That means each section of rope is supporting a quarter of

the total 1000 newton weight i.e. 250 newton. To raise the weight into the air, you have to pull with only a quarter of the force i.e. 250 newton. To make the weight rise 1m, you have to shorten each section of the rope by 1m, so you have to pull the loose end of the rope by 4m. We say a pulley with four wheels and the rope wrapped around like this gives a mechanical advantage of four, which is twice as good as a pulley with two ropes and wheels.

FREE BODY DIAGRAM: (FBD)

Let us draw FBD for various given systems:

- **FBD (1) :** Block of mass M is resting on a frictionless rigid surface shown in figure 4.

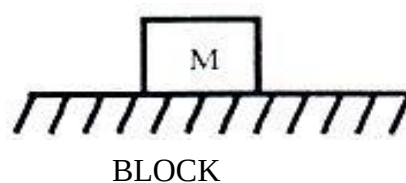
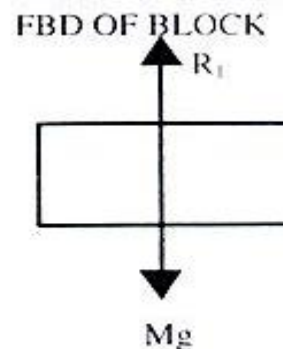


Figure 4



- **FBD (2) :** Draw the free body diagram of the block shown in figure 5.

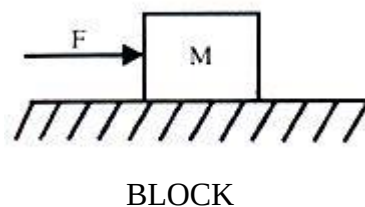
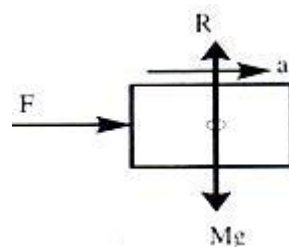


Figure 5



FBD of Block

- **FBD (3) :** If pulley is hanging from a rigid support say roof and masses are connected as shown in figure 6.

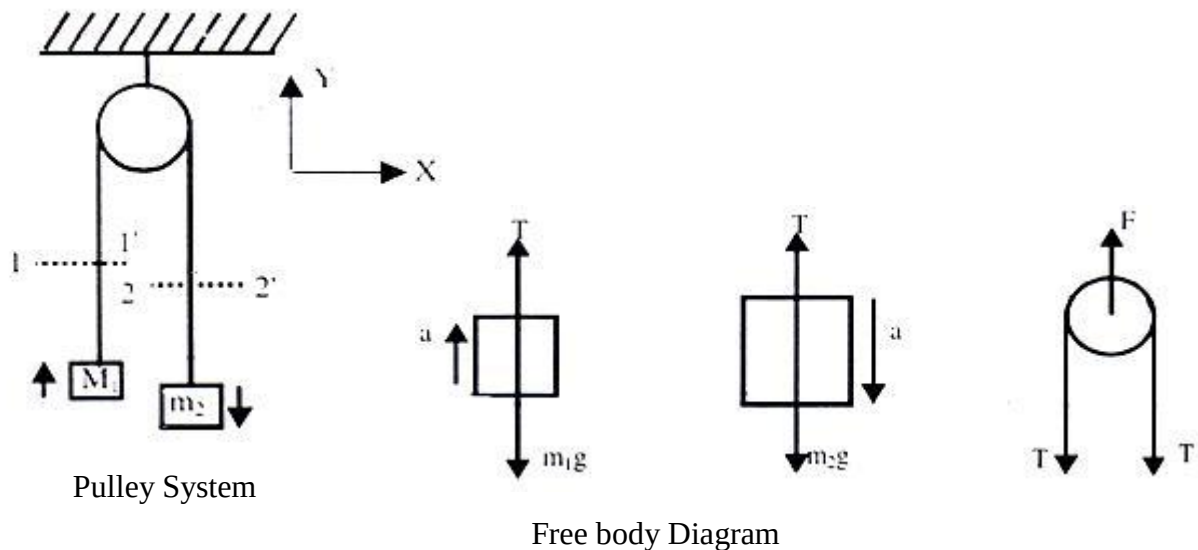
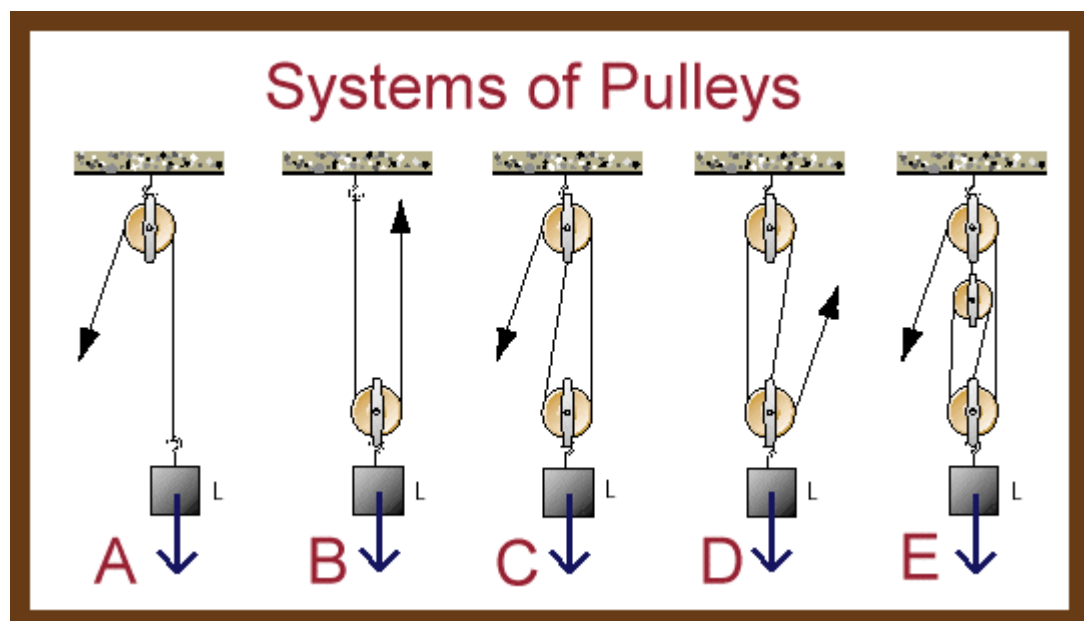


Figure 6

Exercise: Draw the free body diagram for the systems of pulleys shown below.



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STUDY OF FUNDAMENTALS OF KINEMATICS AND KINETICS OF PARTICLES

Aim: To understand the simple dynamic variables involving kinematics, energy and momentum

Theory: i) **KINETICS:** Study of relationship between, motion parameters and the forces.

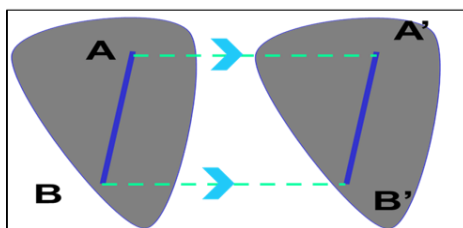
When the body is at rest or moving in a straight line or rotating about an axis, it obeys certain laws. These laws are often known as LAWS OF MOTION. Among all the other laws, Newton's laws are most prominent.

ii) **KINEMATICS:** This is the study of the geometry of motion. It describes the motion of bodies without reference to the forces which cause the motion. It is used to relate position, velocity, acceleration, and time without reference to the cause of the motion.

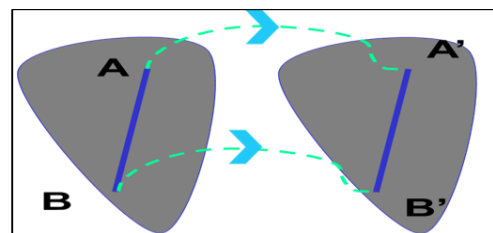
iii) **PLANE OF MOTION:** The plane in which the mass centre of the body moves is defined as the plane of motion.

The plane of motion of rigid body may be divided in to three different categories:

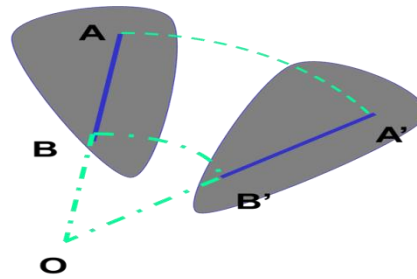
- Translation
- Rotation
- General plane motion



Rectilinear translation



Curvilinear translation



Rotation

iv) **MOTION OF A LIFT:**

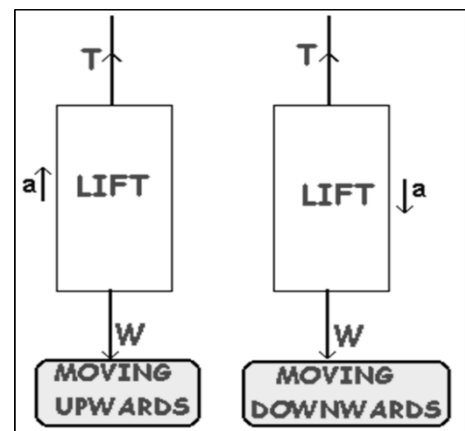
W = Weight of lift + Weight in lift

a = Uniform acceleration of lift

T = Tension in cable

- **For Upward Motion of lift:**

$$F = T - W$$



- **For Downward Motion of lift:**

$$F = W - T$$

- v) **WORK DONE:** Work is measured by the product of force (F) and displacement (S), both being in the same direction.

$$\text{Work done} = \text{Force} * \text{Displacement} = F * S$$

- vi) **ENERGY:** The energy of a body is its capacity of doing work. Various types of energy:

- Mechanical Energy
- Electrical Energy
- Thermal Energy
- Chemical Energy
- Nuclear Energy

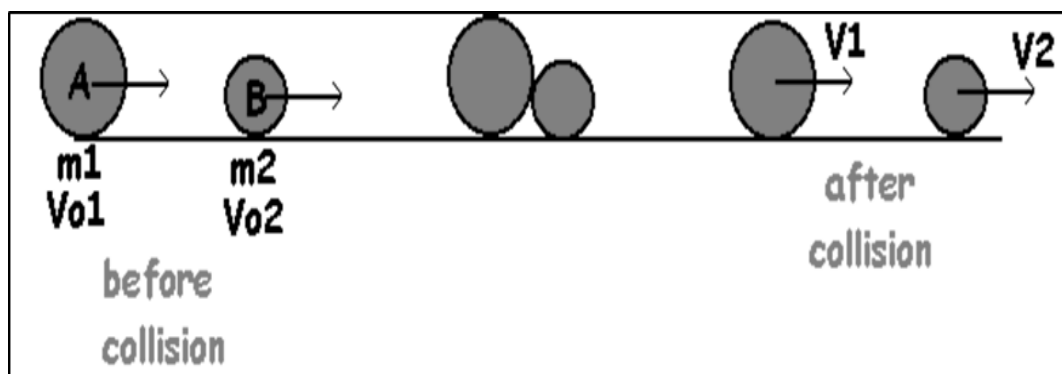
vii) LAW OF CONSERVATION OF MOMENTUM:

When sum of the impulses due to external forces is zero then the momentum of the system remains constant.

Or

“If no external force is acting then total momentum of the system remains unchanged”.

Let's take the case of collision of two balls, A and B



- Before collision,

Momentum of mass $m_a = m_1 v_{01}$

Momentum of mass $m_b = m_2 v_{02}$

Total Momentum = $m_1 v_{01} + m_2 v_{02}$

- After collision,

Momentum of mass $m_a = m_1 v_1$

Momentum of mass $m_b = m_2 v_2$

Total Momentum = $m_1 v_1 + m_2 v_2$

- The law of conservation of momentum requires that, the total momentum before the collision must be equal to the total momentum after the collision.

So,

$$m_1 v_1 + m_2 v_2 = m_1 v_{01} + m_2 v_{02}$$

Exercise: Answer the following Questions.

- 1) Define kinematics and kinetics according to your thought.
- 2) Define Energy and Momentum.
- 3) Explain the law of conservation of momentum.
- 4) Give example of General plane motion.

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